

Sprint 3 Submission - PetAlert (Team Sunrise Inc)

GitHub Repository: <https://github.com/SunriseInc-Capstone/PetAlert>

Team Information and Introduction

Team Name

Sunrise Inc

Project Name

PetAlert

Team Members

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Overview

Problem Context

Pet owners face confusion and lost time when a pet goes missing. Information such as photos, contacts, and identification details is often scattered across different apps or papers, making responses slower and less organized.

Goal

PetAlert aims to provide a cloud-enabled mobile application that securely stores pet profiles, emergency contacts, and alert templates in one centralized location using Firebase services. The system ensures that pet information is persistent, accessible across devices, and quickly retrievable. With a single tap, users can generate and share missing pet alerts via social media, email, SMS, or printable formats, helping reunite pets and families more efficiently.

Requirements

Functional Requirements (User Stories)

- R1 Pet Profile: Create/update pet profiles with photo and details (local storage).
- R2 Emergency Contacts: Add/edit vet, shelter, police contacts with one-tap call/email/SMS.
- R3 Missing Pet Alert: Generate social media post, email/SMS draft, and phone script instantly.
- R4 Regular Update Prompts: Local notifications reminding owners to update info/photos.
- R5 Photo Gallery: Upload/view multiple pet photos locally.
- R6 Recommendations: Static safety tips and search strategies.
- R7 Toxic Foods Infographics: Quick visual guides for dangerous foods.

Non-Functional Requirements

- NF1 Security: All data stored locally; optional OS-level lock (FaceID/Passcode).
- NF2 Reliability: Core features work offline (profiles, contacts, alerts).
- NF3 Usability: ≤3 taps to reach any function; emergency-first button on home.
- NF4 Accessibility: Support large text, high contrast, screen reader labels.
- NF5 Cross-Platform: Runs consistently on iOS 14+ and Android 10+.
- NF6 Accuracy: Generated alerts match stored details exactly.

Design

The app uses a tab-based navigation system:

Home: Prominent “Missing Pet Alert” button.

Pets: Displays all pet profiles with add/edit options.

Pet Detail: Shows stored information and allows generating an alert.

Contacts: Emergency contacts with instant action buttons.

Toxic Foods: Scrollable infographic cards and quick-reference visuals.

Reflection Report

Accomplishments

Sprint 4 focused on migrating the application from local storage to a fully cloud-integrated system. The team successfully transitioned pet profile storage from local JSON and SharedPreferences to Firebase Firestore, ensuring persistent and scalable data management. Firebase Storage was integrated for secure image uploads and retrieval. The microchip number field was added to the pet profile and properly connected to the backend database. Cross-device synchronization was verified using Firebase Authentication sessions. Additionally, UI template updates improved layout consistency, and documentation and testing materials were updated to reflect the new cloud-based architecture. These improvements significantly enhanced the application's scalability and production readiness.

What Went Well

Team coordination during backend migration remained strong. Backend cloud integration was completed smoothly with thorough testing of data storage and retrieval processes. Frontend updates and documentation improvements progressed in parallel, allowing the sprint to maintain steady momentum. Firebase Firestore and Storage integration were implemented without major architectural rewrites, and communication among team members remained consistent and productive.

What Could Be Improved

We tried our best in the provided time. Saying that more structured integration testing between UI and backend services could reduce minor late-stage adjustments. We are planning for that in the next sprint.

Plans for Sprint 5

During this sprint, we will implement the security rules for the backend and continue developing our planned features (emergency contacts, pet alert, photo gallery, toxic food) with the newly improved backend, and if necessary, do more research.

Rubric Reflection

What went well in the presentation?

The Sprint 4 presentation effectively demonstrated the transition to a cloud-based architecture. The team successfully showed pet profile storage using Firebase Firestore, image uploads via Firebase Storage, the integration of the microchip number field, and cross-device synchronization through authenticated accounts. The prototype appeared more scalable and aligned with production-level application standards.

What can be improved in the presentation?

What we can improve in this presentation is to come prepared with a device already connected to the university's Wi-Fi, so internet problems hopefully don't occur again. Unfortunately, we faced a problem with the internet at the beginning of the presentation, but everything worked out because we came prepared with a video demo. We can use a phone hotspot next time to avoid a delay in the demo with the professor or record a video of the demo beforehand again.

Other Notes

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